PERSONALITY IS COMPOSED OF NATURE AND NURTURE

FRANCIS GALTON (1822–1911)

IN CONTEXT

APPROACH
Bio-psychology

BEFORE
1690 British philosopher John Locke proposes that the mind of every child is a tabula rasa, or blank slate, and hence we are all born equal.

1859 Biologist Charles Darwin suggests that all human development is the result of adaptation to the environment.

1890 William James claims that people have genetically inherited individual tendencies, or "instincts."

AFTER
1925 Behaviorist John B. Watson says there is "no such thing as inheritance of capacity, talent, temperament, or mental constitution".

1940s Nazi Germany seeks to create a "master Aryan race" through eugenics.

Personality is composed of elements from **two different sources**.

Nature is that which is inborn and inherited, and…

Nurture is that which is experienced from birth onward.

We can improve our skills and abilities through **training and learning**, but…

…nature **sets the limits** to how far we can develop our talents.

Nature and nurture both play a part, but **nature is the determining factor**.

Francis Galton counted many gifted individuals among his relatives, including the evolutionary biologist Charles Darwin. So it's not surprising that Galton was interested in the extent to which abilities are either inborn or learned. He was the first person to identify "nature" and "nurture" as two separate influences whose effects could be measured and compared, maintaining that these two elements alone were responsible for determining personality. In 1869, he used his own family tree, as well as those of "judges, statesmen,

See also: John B. Watson 66–71 ▪ Zing-Yang Kuo 75 ▪ G. Stanley Hall 46–47 ▪ Eleanor E. Maccoby 284–85 ▪ Raymond Cattell 314–15

> Characteristics cling to families.
> **Francis Galton**

commanders, scientists, literary men… diviners, oarsmen, and wrestlers," to research inherited traits for his book *Hereditary Genius*. As predicted, he found more highly talented individuals in certain families than among the general population. However, he could not safely attribute this to nature alone, as there were also conferred benefits from growing up in a privileged home environment. Galton himself grew up in a wealthy household with access to unusually good educational resources.

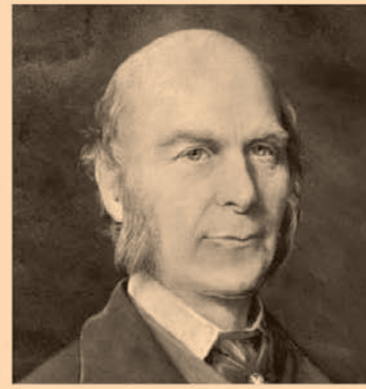
A necessary balance

Galton proposed a number of other studies, including the first large survey by questionnaire, which was sent out to members of the Royal Society to inquire about their interests and affiliations. Publishing his results in *English Men of Science*, he claimed that where nature and nurture are forced to compete, nature triumphs. External influences can make an impression, he says, but nothing can "efface the deeper marks of individual character." However, he insists that both nature and nurture are essential in forming personality, since even the highest natural endowments may be "starved by defective nurture." Intelligence, he says, is inherited, but must be fostered through education.

In 1875, Galton undertook a study of 159 pairs of twins. He found that they did not follow the "normal" distribution of similarity between siblings, in which they are moderately alike, but were always extremely similar or extremely dissimilar. What really surprised him was that the degree of similarity never changed over time. He had anticipated that a shared upbringing would lessen dissimilarity between twins as they grew up, but found that this was not the case. Nurture seemed to play no role at all.

The "nature–nurture debate" continues to this day. Some people have favored Galton's theories, including his notion—now known as eugenics—that people could be "bred" like horses to promote certain characteristics. Others have preferred to believe that every baby is a tabula rasa, or "blank slate," and we are all born equal. Most psychologists today recognize that nature and nurture are both crucially important in human development, and interact in complex ways. ▪

Galton's study of twins looked for resemblances in many ways, including height, weight, hair and eye color, and disposition. Handwriting was the only aspect in which twins always differed.



Francis Galton

Sir Francis Galton was a polymath who wrote prolifically on many subjects, including anthropology, criminology (classifying fingerprints), geography, meteorology, biology, and psychology. Born in Birmingham, England, into a wealthy Quaker family, he was a child prodigy, able to read from the age of two. He studied medicine in London and Birmingham, then mathematics at Cambridge, but his study was cut short by a mental breakdown, worsened by his father's death in 1844.

Galton turned to traveling and inventing. His marriage in 1853 to Louisa Jane Butler lasted 43 years, but was childless. He devoted his life to measuring physical and psychological characteristics, devising mental tests, and writing. He received many awards and honors in recognition of his numerous achievements, including several honorary degrees and a knighthood.

Key works

1869 *Hereditary Genius*
1874 *English Men of Science: Their Nature and Nurture*
1875 *The History of Twins*